

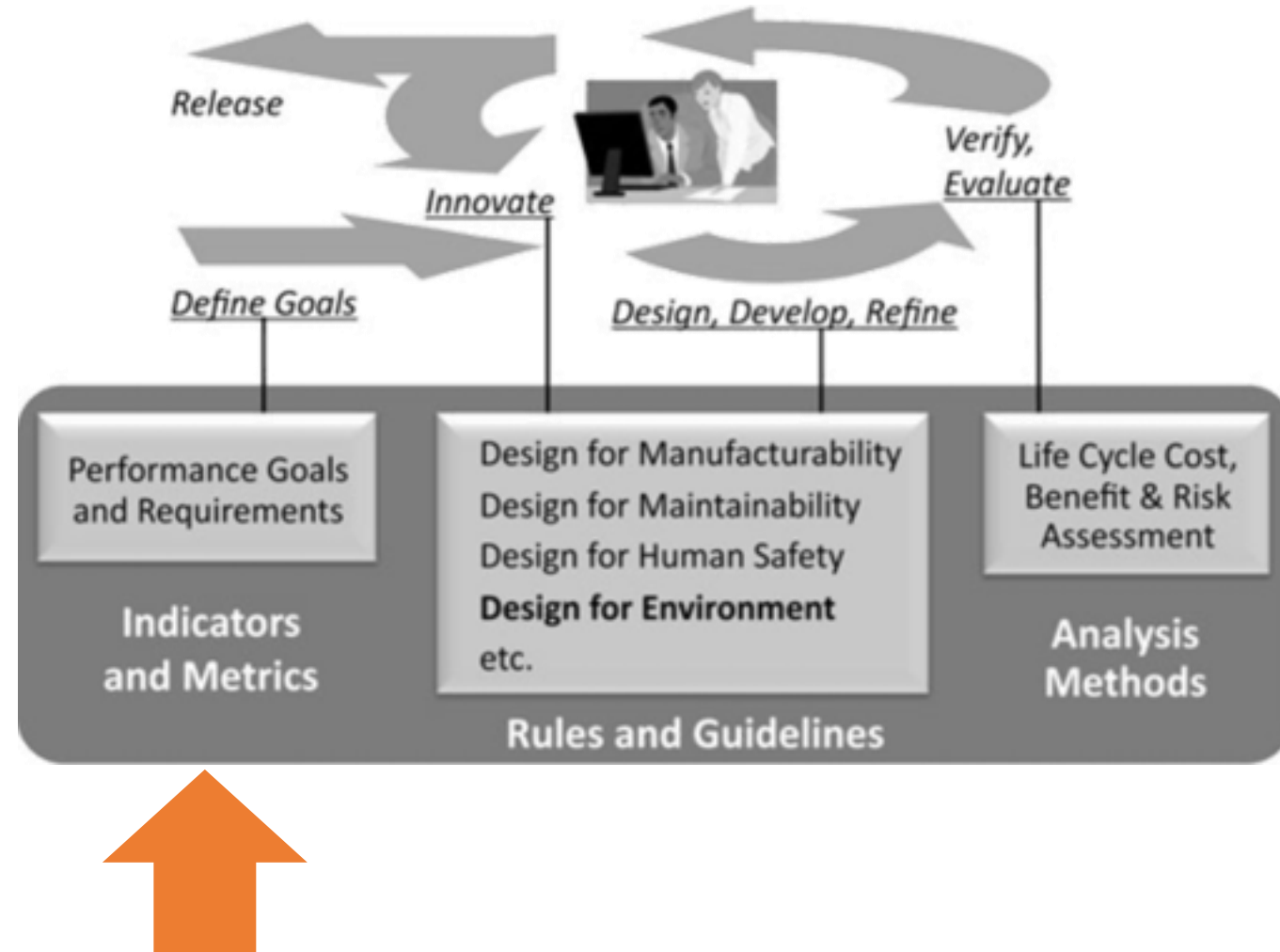
1.4 Performance Indicators and Metrics

Dr Peter Kovalsky

Recap: The Seven Principles of DFE

1. Embed **life-cycle thinking** into the engineering process
2. Evaluate the **material and energy efficiency** and effectiveness of the overall system
3. Select appropriate **metrics** to represent product life-cycle performance
4. Maintain and apply a portfolio of systematic **design strategies**
5. Use **analysis methods** to evaluate design performance and understand trade-offs
6. Provide **software** capabilities to facilitate the application of DFE practices
7. Seek **inspiration from nature** for the design of products and systems

First stage of product/service development...





Definitions

- **Performance Indicators and Metrics:** Guides for product development, relevant and meaningful to those on the design team and other stakeholders (customers, etc.).
- **In a DFE context:** Accounting systems that explicitly recognise and track environmental costs and benefits.
- It is common to distinguish between
 - *Indicators:* defines what is to be measured
 - *Metrics:* defines how it will be measured.
- **KPI's:** Key performance indicators

Examples of DFE indicators/metrics:

- Resource utilisation
 - e.g., total energy consumed during the product life cycle
- Atmospheric emission
 - e.g., greenhouse gases and ozone-depleting substances released per unit of production
- Waste minimization
 - e.g., percent of product materials recovered at end-of-life
- Toxic substance use
 - e.g., total kg of solvents purchased per unit of production





Importance of indicators/ metrics:

- Helps set initial product goals and objectives in the concept phase of product development
- Improves stakeholder communication
- Assigns accountability
- Motivates internal process improvement
- “You can’t manage what you can’t measure”

Application in DFE

Selecting indicators and metrics:

- Who is the intended audience and why?
- What do the performance indicators/metrics signify to that audience? Clear? Relevant?
- Will they withstand the test of changing market conditions over an extended time period?

For product life-cycle management, three important facets include:

- **Measuring** continuous **improvement** in performance over time
- **Communicating** with customers and stakeholders about **product advantages**
- **Benchmarking performance** against comparable products and processes



Eco efficiency: “do more with less”

The ratio of product/service value delivered to the environmental burdens of product or service creation or consumption

- Example: resource productivity
e.g., number of vehicles produced per kilowatt-hour of energy used, gallon of water used, or ton of greenhouse gas emitted.



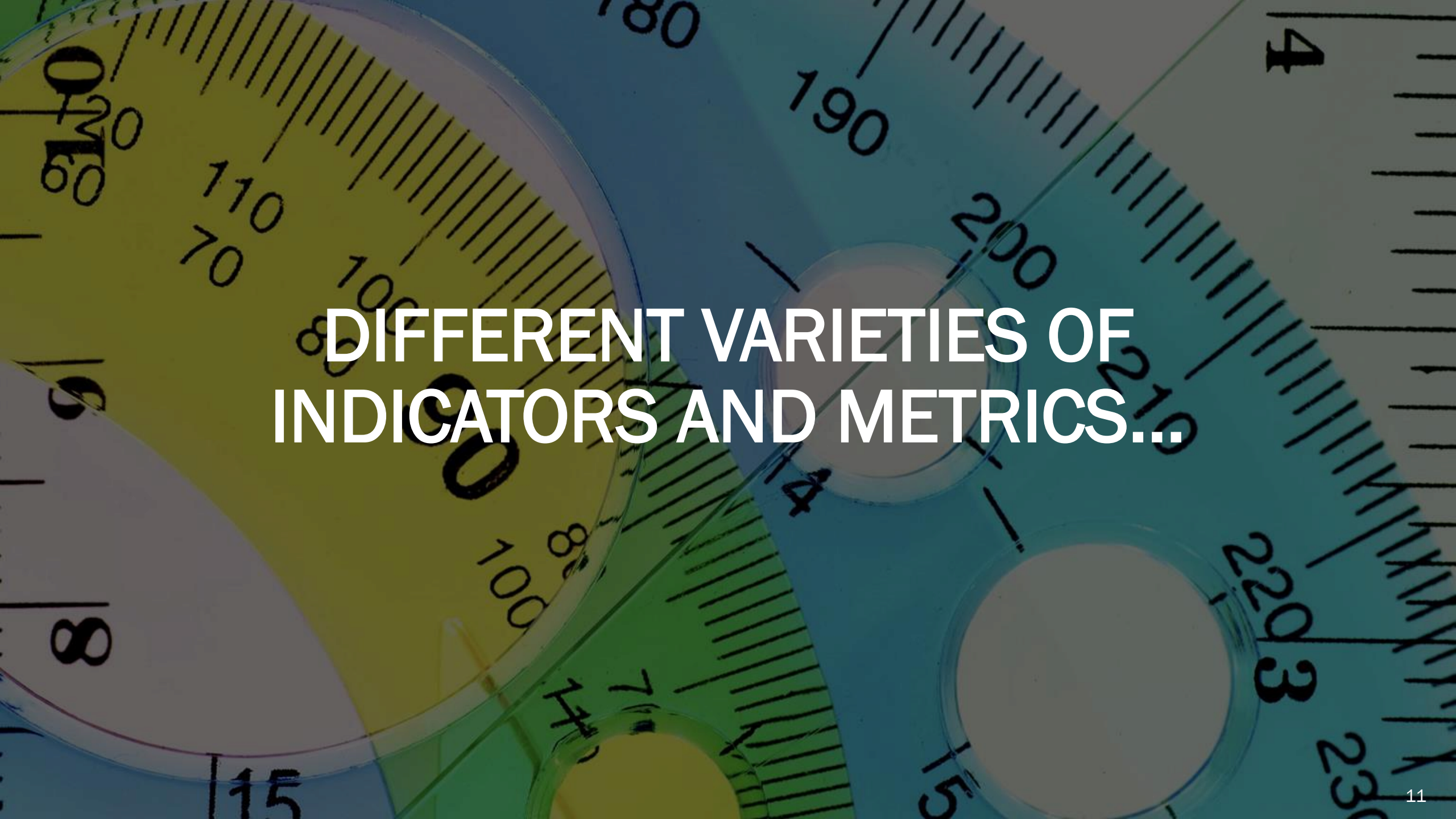
The indicators and metrics should be...

1. Relevant to the interests of the intended audiences (stakeholders, management, etc.)
2. Comprehensible, transparent and clear in meaning to the intended audiences
3. Objective in terms of measurement techniques and verifiability
4. Effective for supporting benchmarking and monitoring over time, as well as deciding how to improve performance
5. Comprehensive in providing an overall evaluation of the company's products and services
6. Consistent across different sites or facilities, using appropriate normalization methods
7. Practical in allowing cost-effective, non-burdensome implementation and building on existing data collection where possible

Example: Procter & Gamble (P&G)

- US consumer goods company specializing in hygiene products, e.g. soap
- Hence it focuses its sustainability efforts in:
 - *Water—since 85% of P&G products involve household water use, the company has focused on how it can enhance both water conservation and water quality.*
 - *Health and hygiene—the company has sought to enhance the global contributions of its products to cleanliness, sanitation, health care, infant care, and education*
- Possible metrics: kg product produced /kg of water used in production.
kg product used/kg water used during product use.



The background of the slide features a collage of overlapping circular gauges or dials. These gauges have various scales and numbers, including 0, 20, 60, 70, 80, 100, 110, 120, 130, 140, 150, 160, 170, 180, 190, 200, 210, 220, 230, 240, 250, 260, 270, 280, 290, 300, 310, 320, 330, 340, 350, 360, 370, 380, 390, 400, 410, 420, 430, 440, 450, 460, 470, 480, 490, 500, 510, 520, 530, 540, 550, 560, 570, 580, 590, 600, 610, 620, 630, 640, 650, 660, 670, 680, 690, 700, 710, 720, 730, 740, 750, 760, 770, 780, 790, 800, 810, 820, 830, 840, 850, 860, 870, 880, 890, 900, 910, 920, 930, 940, 950, 960, 970, 980, 990, 1000. The gauges are in various colors, including yellow, green, blue, and red. The text "DIFFERENT VARIETIES OF INDICATORS AND METRICS..." is overlaid in the center in a bold, white, sans-serif font.

DIFFERENT VARIETIES OF INDICATORS AND METRICS...

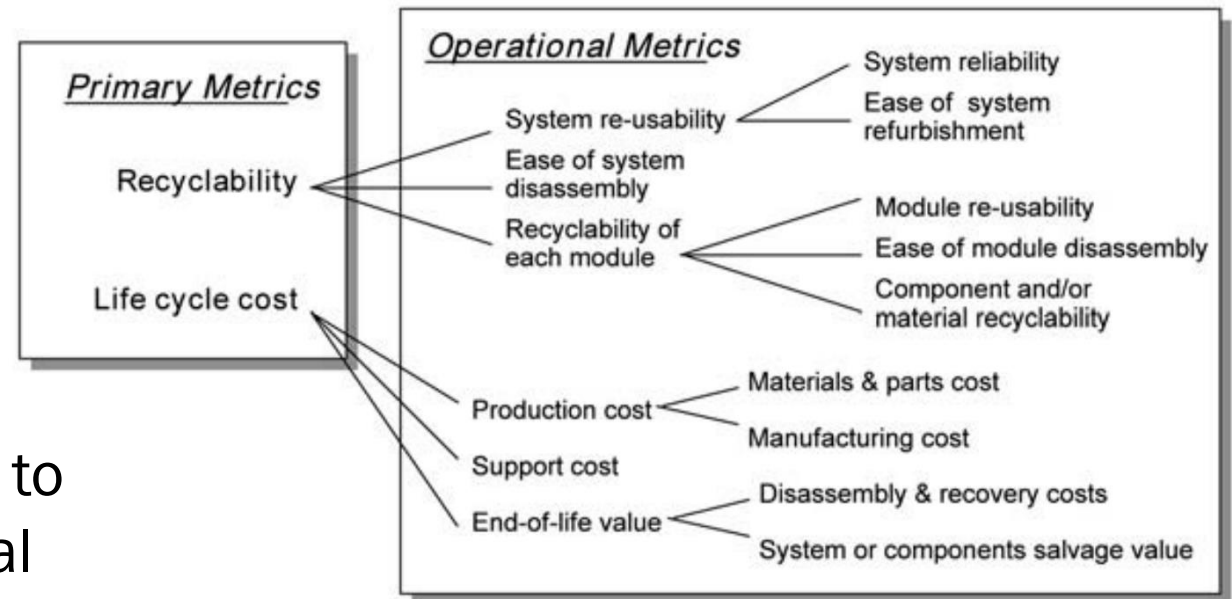


Leading vs Lagging indicators

- **Leading** indicators measure internal business process improvements that are expected to **drive future performance**
- **Lagging** indicators, also referred to as **outcome** indicators, measure the actual results (e.g., emission reductions) attributable to operational improvements
 - *Pros: meaningful to stakeholders*
 - *Cons: retrospective (do not provide foresight about future performance)*
- It's often good to couple leading and lagging indicators together:
 - *Example: Motorola reduced occupational injuries (lagging indicator) by tracking the cause of wooden pallet defects (leading indicator)*

Primary and Operational metrics

- **Primary** metrics are driven either by a fundamental customer need or by important internal constraints
- **Operational** metrics represent observable and controllable measures associated with a product or process
- Primary metrics generally need to be decomposed into operational metrics



A close-up photograph of a hand placing a coin on top of a stack of several other coins. The coins are gold-colored and the background is a soft, out-of-focus blue.

Primary Indicator and metric examples

■ Indicator: energy usage

- **Metrics:** *Total energy consumed during the product life cycle*
- *Renewable energy consumed during the life cycle*
- *Power used during operation (e.g., for electrical products)*

■ Indicator: Economics

- **Metrics:** *Average life-cycle cost incurred by the manufacturer*
- *Purchase and operating cost incurred by the customer*
- *Cost savings associated with design improvements*

■ Indicator: Value creation

- **Metrics:** *Utilization of renewable resources*
- *Avoidance of waste or pollution*
- *Human health and safety improvement*
- *Enhancement in environmental quality or biodiversity*
- *Enhanced economic development and job creation*

Qualitative vs quantitative

- **Qualitative** metrics are those that rely upon observation and judgment, e.g. brand perception
 - *Pros: they impose a relatively small data collection burden and are easy to implement*
 - *Cons: they incorporate subjective information and are therefore difficult to validate*
- **Quantitative** metrics are those that rely upon empirical data and derive numerical results, e.g. kg of waste
 - *Pros: they are objective, meaningful, and verifiable*
 - *Cons: the required data may be burdensome to gather or simply unavailable*



Absolute vs. relative

- **Absolute** metrics are those that are defined with respect to a fixed measurement scale, e.g. total annual hazardous waste generated.
 - *Pros: reflects actual environmental impacts rather than efficiency of production*
 - *Cons: may lead to inappropriate comparisons*
- **Relative** metrics are those that are defined with respect to another metric or variable. This occurs where flows of resources or emissions are normalized by total sales, time, or production volume, e.g. total hazardous waste generated per unit produced
 - *Pros: generally less biased*
 - *Cons: reflect efficiency of production rather than actual environmental impacts*



Source vs Impact

Source vs Impact

- **Source metrics:** address the root causes of environmental consequences associated with an organization's activities. E.g. quantity of toxic materials released at a given site
 - *Pros: they are both readily observable and controllable*
 - *Cons: indirect indicator of potential impacts and can overlook important impact pathways*
- **Impact metrics** are those that address the actual environmental consequences which may result from an organization's activities
 - *Pros: directly addressing the impacts of concern*
 - *Cons: challenging to develop techniques that address them due to technical and statistical uncertainties*
- In practice, the most efficient means of performance measurement is to select company-specific indicators = typically source-oriented indicators.

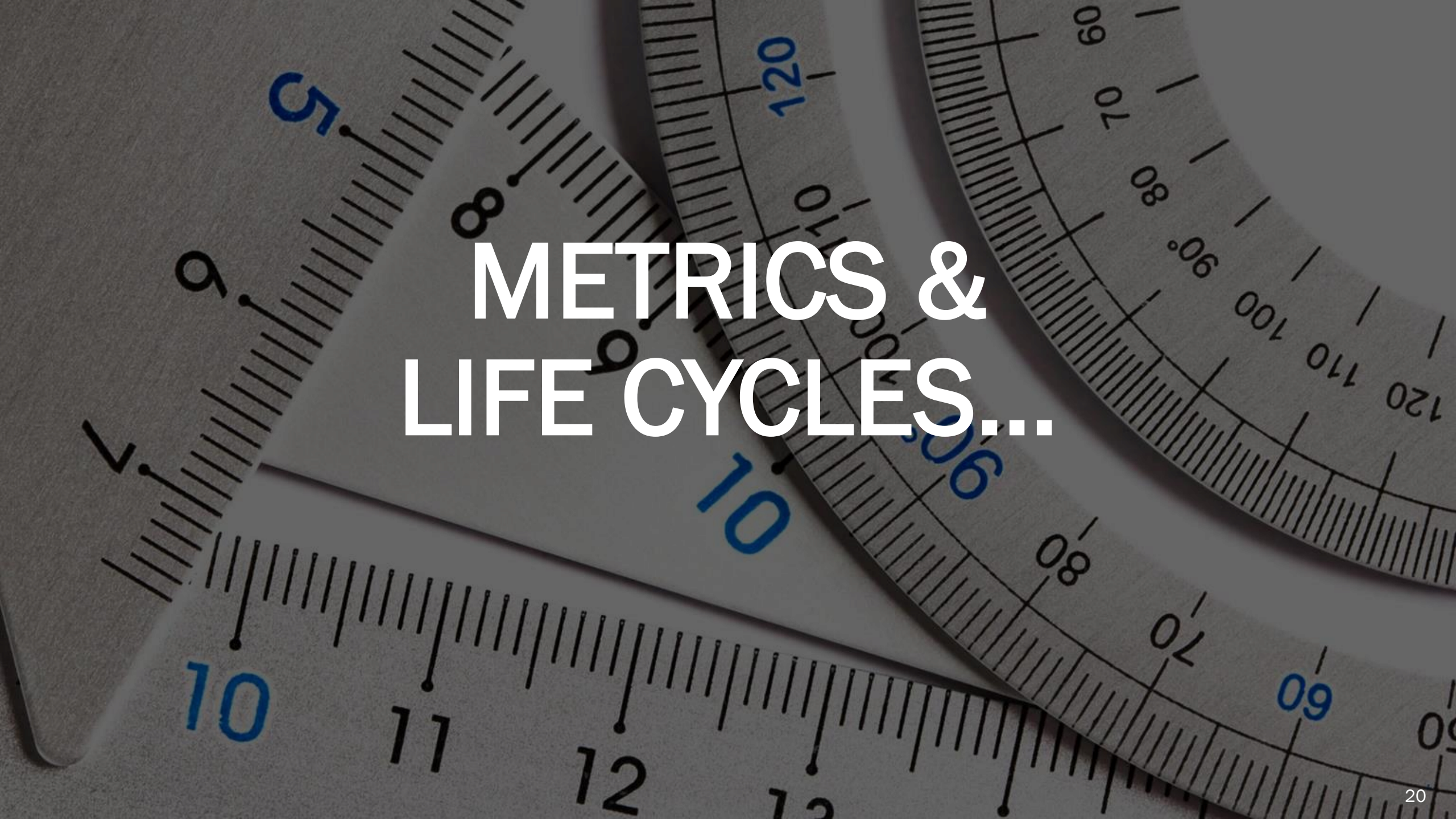
Example: airborne emissions

- ***Quantitative relative source*** metric which allows companies and regulatory agencies to establish clear targets for improvement
- Metric: waste ratio (eco-efficiency)

$$\text{Waste ratio} = \frac{\text{Waste}}{\text{Product} + \text{By-product} + \text{Waste}}$$

- In order to minimize this ratio, engineers can either convert waste to useful by-products or reduce the waste generated.





METRICS & LIFE CYCLES...

Some metrics are only applicable to certain parts of a product life cycle:

	Materials	Fabrication	Transport	End Use	Disposal
Energy usage ↓	★	★	★	★	★
Water usage ↓	★	★		★	
Source volume ↓	★	★	★	★	★
Recycling & reuse ↑		★		★	★
Waste & emissions ↓	★	★	★	★	★
Recycled materials ↑	★	★			

TABLE 7.2 Relevance of Environmental Performance Metrics to Various Product Life-Cycle Stages

Interactions between metrics:

- Engineers must understand the synergies and tradeoffs among environmental performance attributes
- Environmental metrics can relate to cost, performance, and customer satisfaction.
- E.g. synergies between lifecycle cost reduction and a number of environmental performance metrics (water use, energy use, emissions reduction etc)

	Performance	Cost	Safety	Convenience	Aesthetics
Energy usage ↓		★			
Water usage ↓		★		★	
Source volume ↓	?	★	?	★	
Recycling & reuse ↑		★		?	
Waste & emissions ↓	★	★	★	★	
Recycled materials ↑		?			?

TABLE 7.3 Example of Interactions among Environmental Quality and Other Customer Benefits for a Consumer Product

Points to remember for your DFE report...

Text book: Chapter 7

Design for Environment: A Guide to Sustainable Product Development
Joseph Fiksel

Goals	Examples of Metrics	Examples of Specific Objectives
Reduce or eliminate waste	• lb. of emissions over the life cycle • % of product weight disposed of in landfills	• Reduce life-cycle emissions by 30% annually • Reduce solid waste disposed of to 1 lb. per product unit
Develop “green” recyclable products	• % of product weight recovered & recycled • toxic or hazardous constituents	• Achieve 95% recycling • Eliminate use of brominated compounds
Reduce life-cycle cost of product	• manufacturing cost • distribution and support cost • end-of-life cost	• Reduce total life-cycle cost to \$7,500 per product unit • Reduce end-of-life cost (or increase value) by 20%
Cost of ownership for customers	• annualized purchase and operating cost (\$)	• Must be less than \$500 per year
Conserve energy consumption over the life cycle	• total energy (BTU) to produce one unit • average power use (for devices)	• Reduce to 1000 BTU • Reduce by 10% annually • Power less than 30 watts
Conserve natural resources by raising the recycled content	• % by weight of product materials from sources	• Achieve 20% or greater total recycled content • Achieve 30% recycled plastics

TABLE 7.1 Selected Environmental Goals and Corresponding Metrics

Summary

- Environmental sustainability has become a key factor for businesses in long-term competitiveness
- Performance indicators and metrics are needed for tracking environmental costs and benefits: “You can’t manage what you can’t measure”
- Life cycle thinking should be considered when applying indicators and metrics
- Metric interactions/ synergies can occur
- In the long run, as sustainability becomes integrated into a company’s operations, the distinction between environmental goals and business goals should vanish

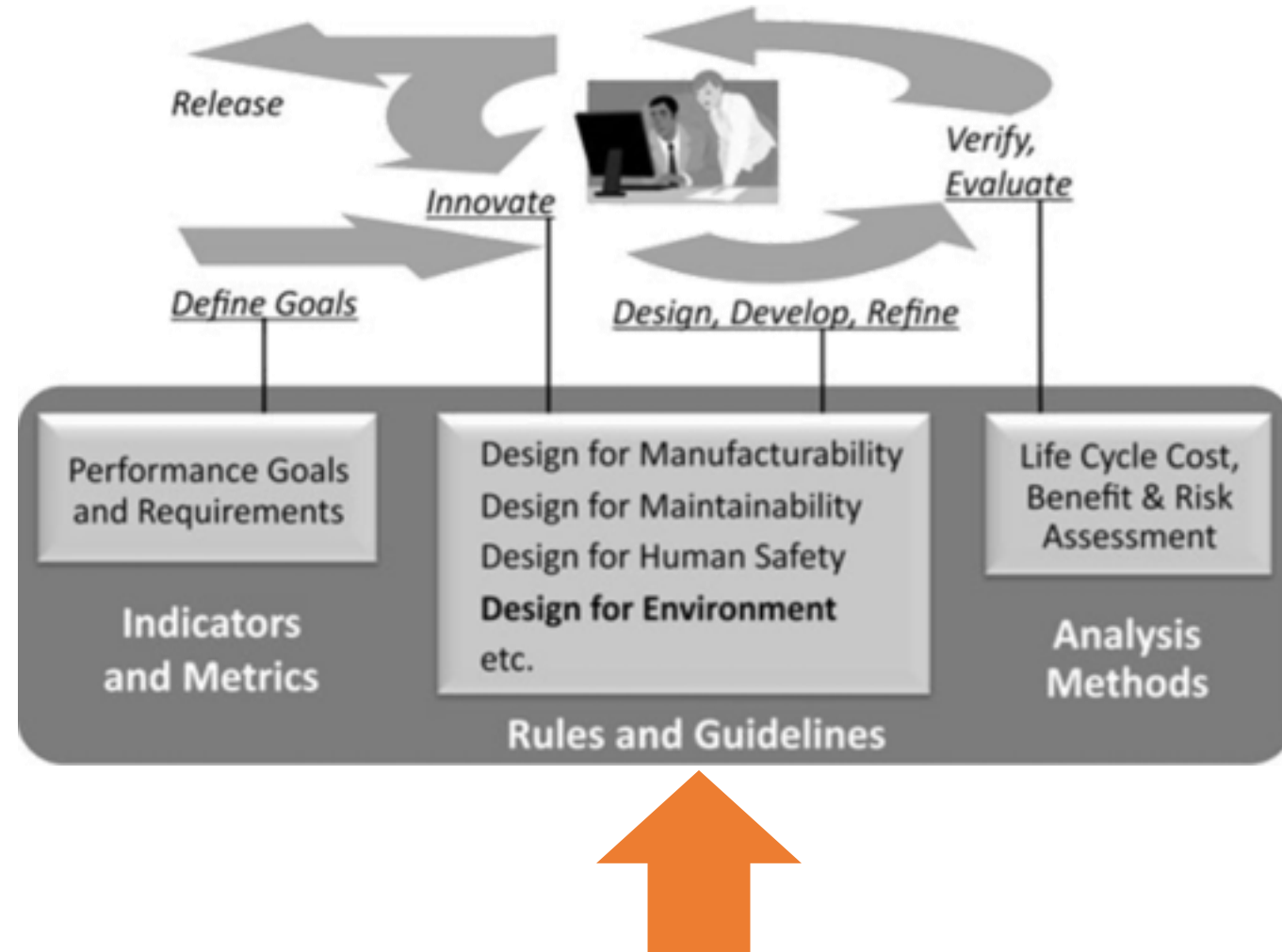
1.5 Design for Dematerialisation & Detoxification

Dr Tim Walmsley

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Second stage of product/service development...





Guideline definitions:

- Guidelines help refine the product development process...
 - ***Prescriptive guidelines** are definite statements about what designers should or should not do, and are sometimes called design rules.*
 - ***Suggestive guidelines** represent accumulated knowledge, including best practices and lessons learned, but they are not strict rules.*
- DFE guidelines are often suggestive guidelines

Importance of guidelines:

- They encourage consistency among different development teams e.g., standard material labeling schemes.
- They promote the accumulation of knowledge over successive design cycles and allow that knowledge to be preserved and passed down.
- They lead to a more systematic design process that is less dependent upon the biases of individual designers.
- They expand the scope of issues considered during design, allowing the team to anticipate downstream pitfalls or constraints that they may have ignored.

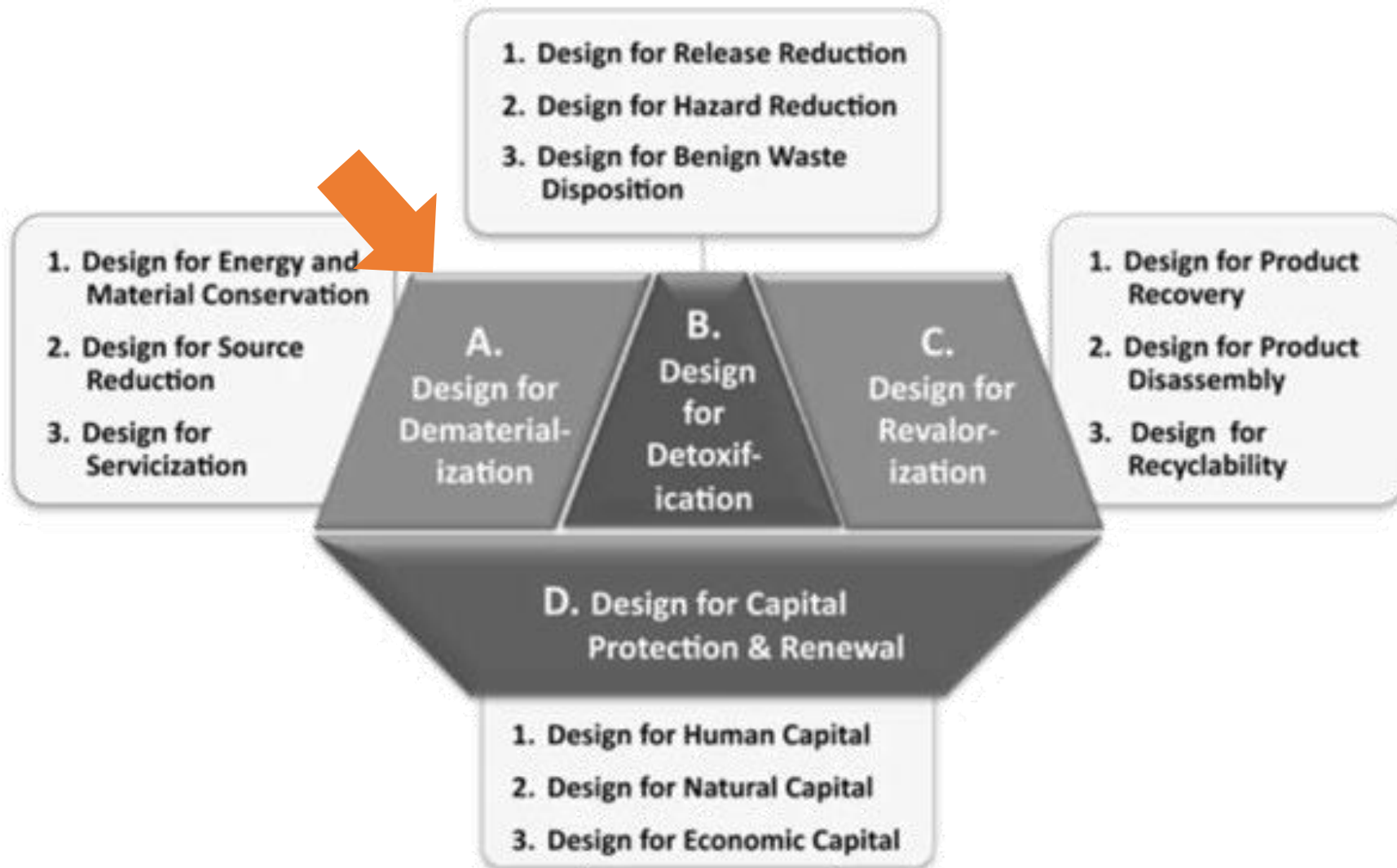


Four principal guideline strategies:

- **Material** and **energy** used in product manufacture and its associated processes throughout their life cycle.
- Design for **detoxification** seeks to reduce or eliminate the **toxic, hazardous**, or otherwise harmful characteristics of a product and its associated processes.
- Design for **revalorization** seeks to **recover, recycle**, or otherwise **reuse** the residual materials and energy that are generated at each stage of the product life cycle.
- Design for **capital protection** and renewal seeks to ensure the **safety, integrity, vitality, productivity**, and **continuity** of the human, natural, and economic resources that are needed to sustain the product life cycle

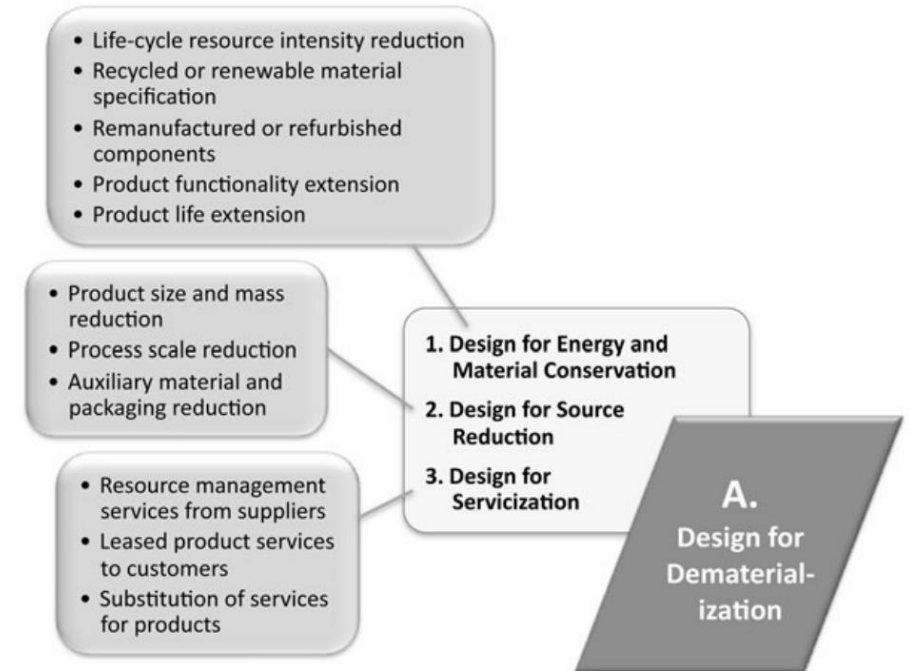


Design for Dematerialization ...



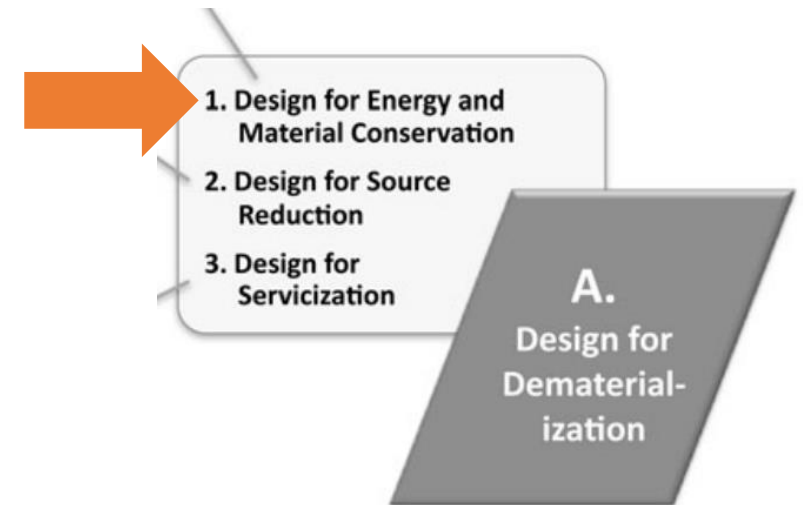
Design for Dematerialization

- **Dematerialization**
definition: the reduction of material throughput in an economic system
- Material flow is a fundamental driver of energy use, water use, greenhouse gas emissions, and most other environmental indicators
- The more we use, the more we waste



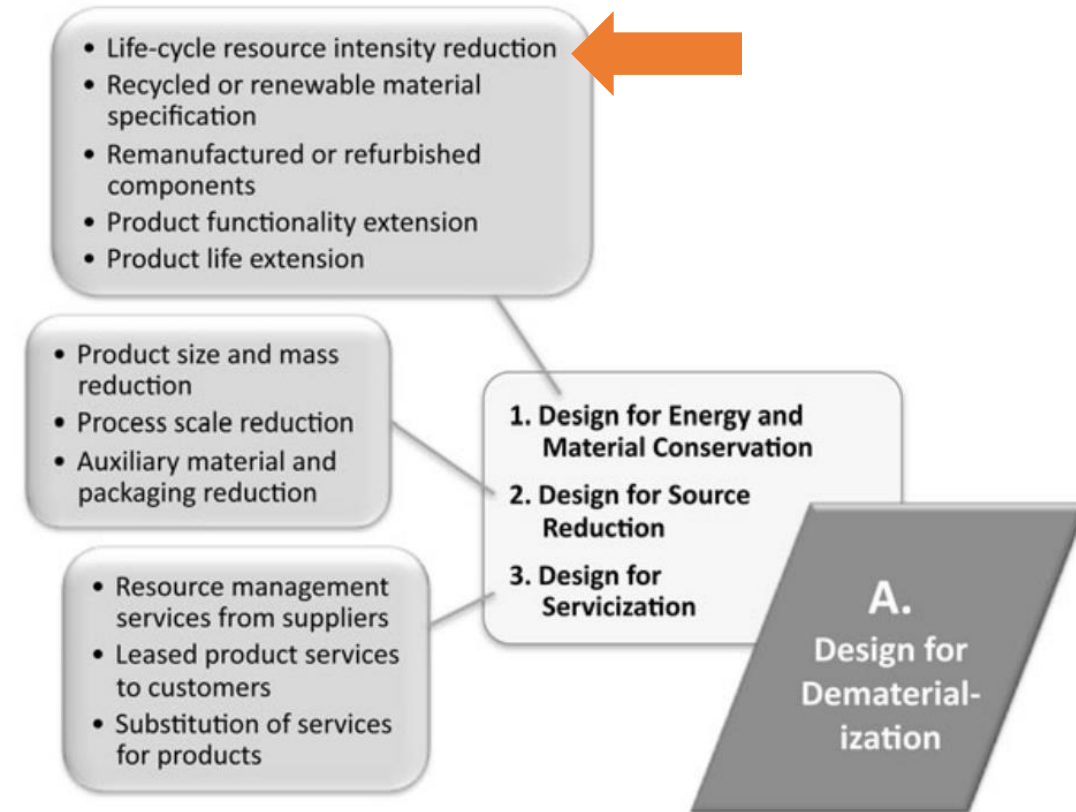
Technique: A.1 Design for Energy and Material Conservation

- Most direct method
- Synergistic with business goals (reductions in operating costs)
- Energy conservation that reduces fossil fuel use will also reduce greenhouse gas emissions

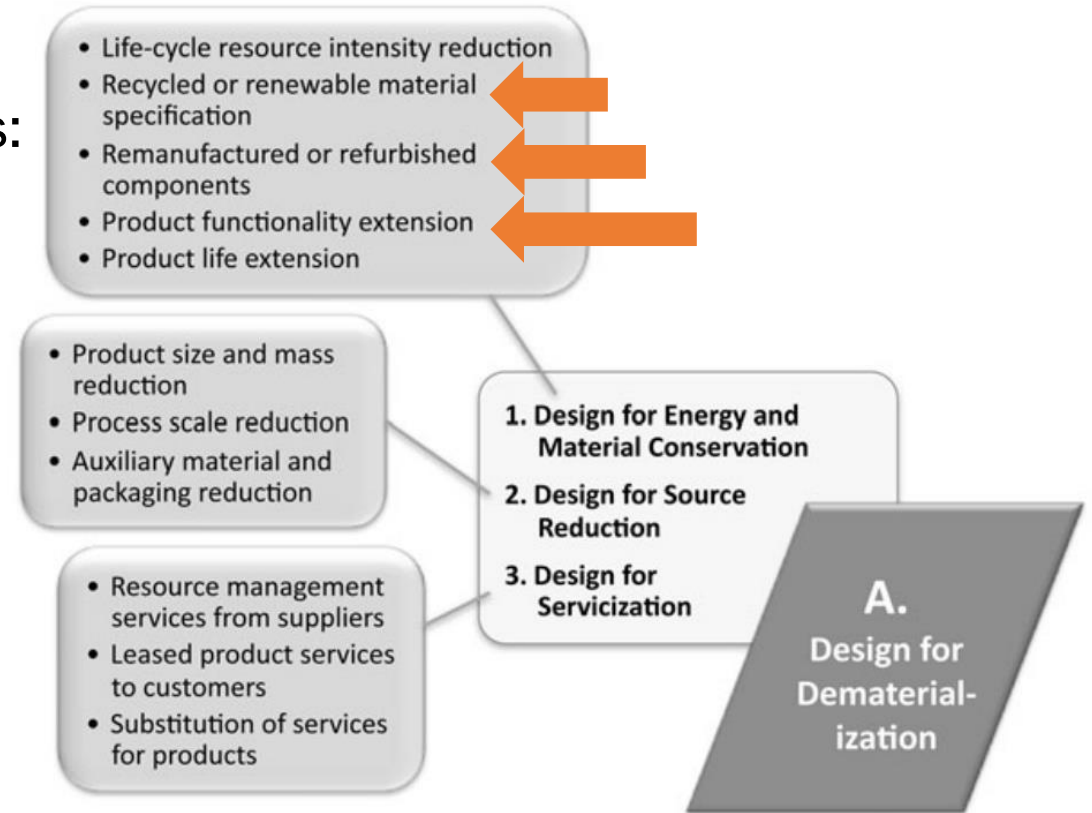


Life-Cycle Resource Intensity Reduction

1. Reduce the procurement footprint.
 - *Example: Walmart scores suppliers on sustainability of packaging*
2. Reduce the operational resource footprint
 - *Example: tightening up on heating, cooling, and lighting system*
3. Reduce transportation requirements
 - *Example: Reduce the total transportation distance by shipping outsourced modules directly from the supplier to the final customer*
4. Reduce end-use power consumption
 - *Example: Power management features on laptops*
5. Reduce end-use material consumption
 - *Example: having a rechargeable battery in an electrical device instead of using disposable batteries*

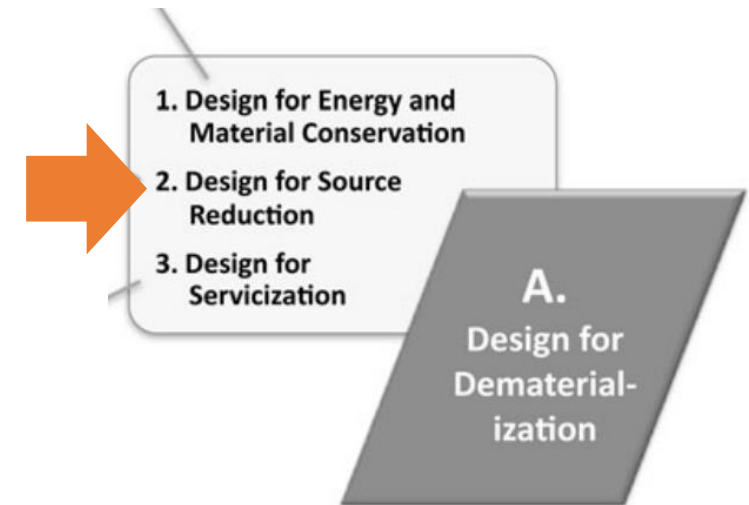


- Recycled or Renewable Material Specification:
 - *Example: Use of natural materials (e.g. flax) for car interiors*
- Remanufactured or Refurbished Components:
 - *Example: Caterpillar remanufacturing business takes end-of-life engine and mining components and remanufactures them to fit engineering updates*
- Product Functionality Extension:
 - *Examples: All-in-one copiers that double as printers, scanners, and fax machines*
- Product Life Extension:
 - *Technological obsolescence issue e.g. laptops*



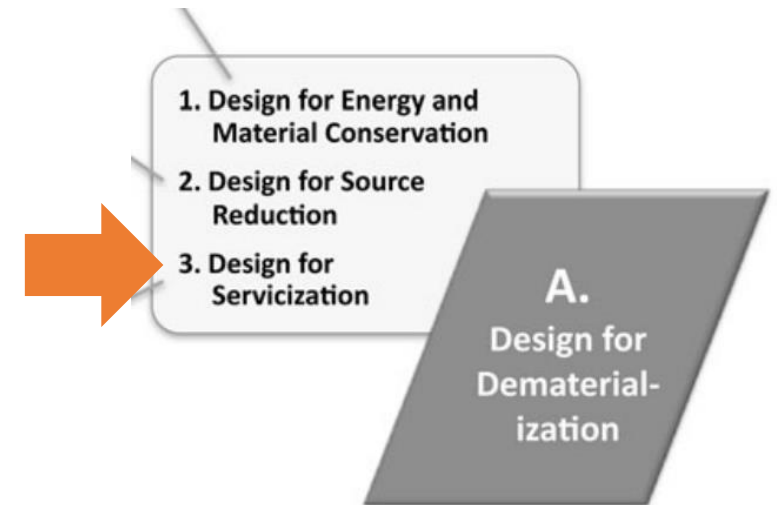
Technique: A.2 Design for Source Reduction

- Reducing the mass of a product is the surest and most direct way to achieve waste reduction
- Reduce the product's physical dimensions
 - *Example: Electronics mass steadily decreasing*
- Reduce the mass of key components
 - *Example: new battery technologies have enabled smaller and lighter battery packs*
- Specify lighter-weight materials as substitutes
 - *Example: composite materials with superior strength-to-weight ratios (although this property may present recyclability challenges)*



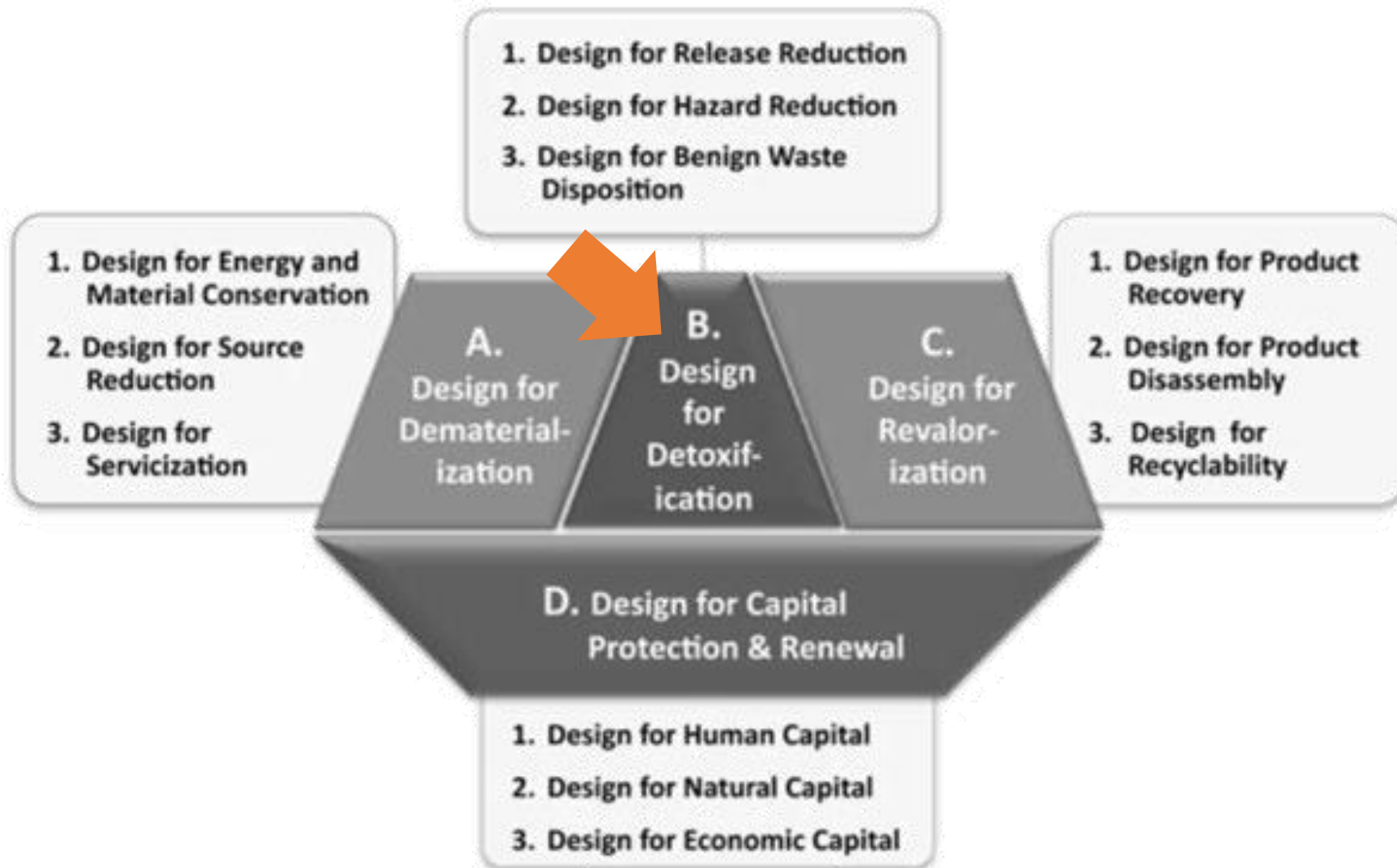
Technique: A.3 Design for Servicization

- The most radical approach to dematerialization is to eliminate products altogether and provide services instead
- Examples:
 - *Mobile services such as lime scooters*
 - *Xerox: fixed lease cost, customers receive technical support of the product until they are ready to upgrade, and manufacturers can then recover the used product and maximize its residual value*



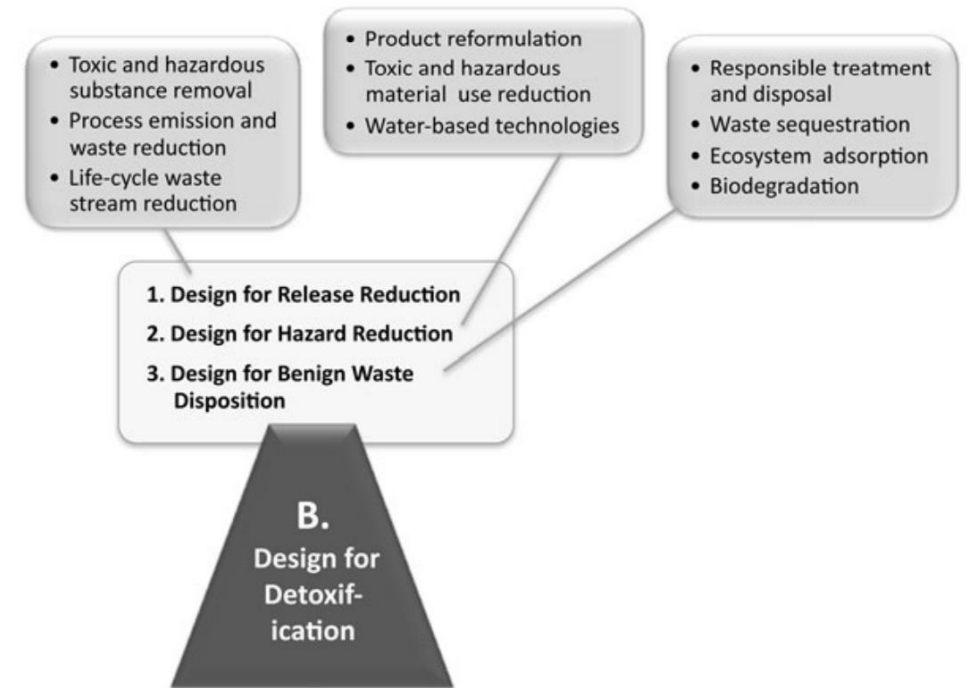
Design for Detoxification...





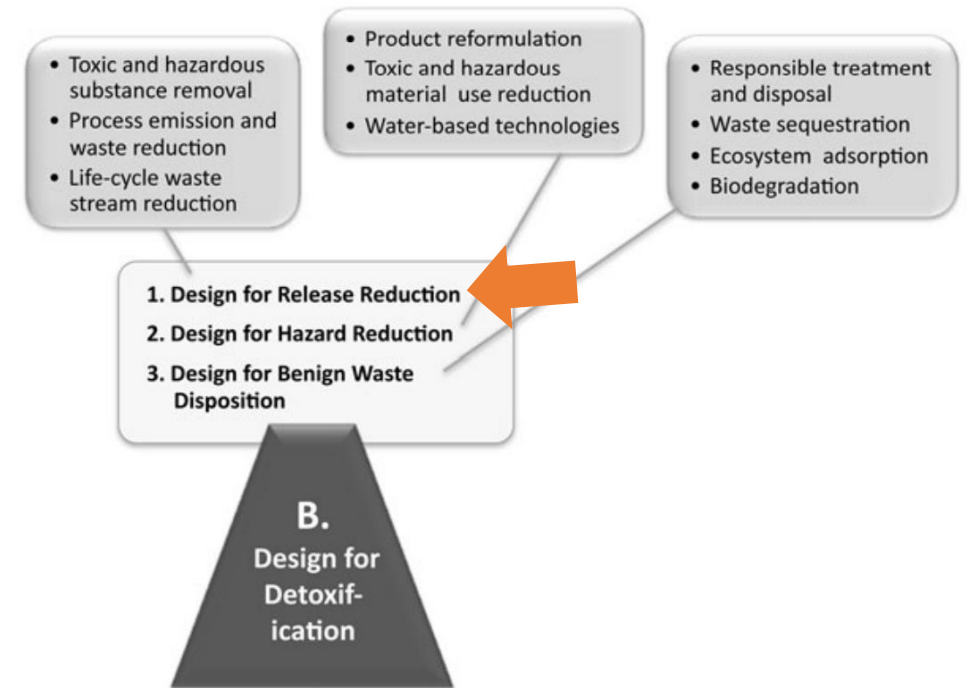
Design for Detoxification

- **Detoxification definition:** the prevention or reduction of adverse human or ecological effects associated with materials use.
- Simply reducing the mass of materials is insufficient.
- Some materials released into the environment can have significant adverse human health and environmental impacts even at very low levels
 - e.g. lead, mercury, and dioxin.



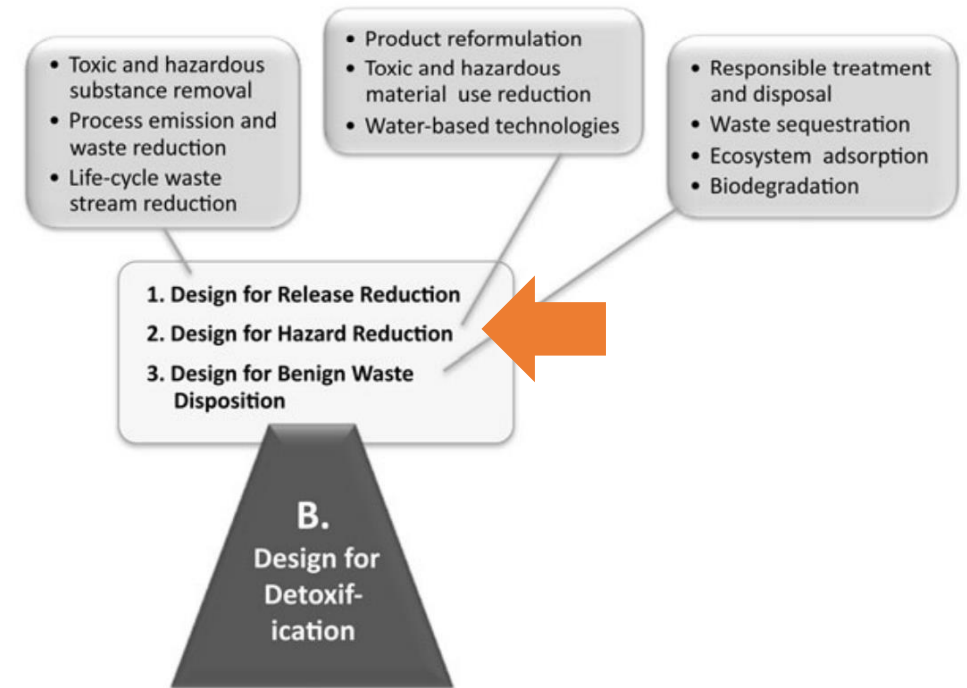
Technique: B.1 Design for Release Reduction

- Toxic and Hazardous Substance Removal
 - *Example: Airborne emissions may be partially removed through pollution control devices, such as oxidizers, scrubbers, filters, or electrostatic precipitators.*
- Process Emission and Waste Reduction
 - *Example: Reduce excess inventory to prevent product spoilage and waste*
- Life-Cycle Waste Stream Reduction
 - *Example: battery recycling*



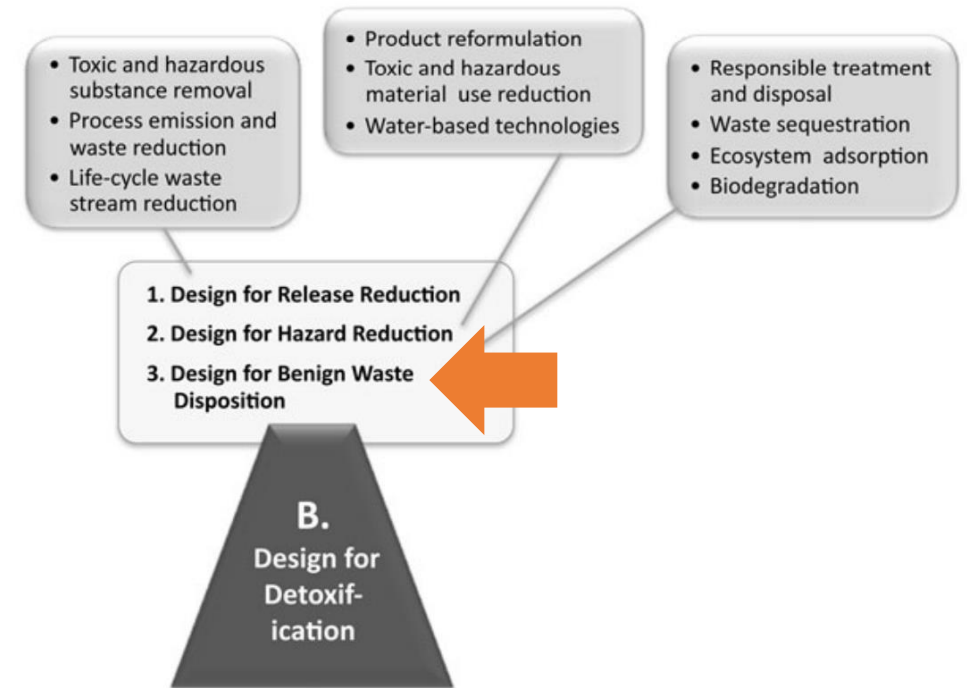
Technique: B.2 Design for Hazard Reduction

- In some cases, there are unintended adverse consequences
 - *Example: replacing lead with MBTE in gasoline resulted in widespread contamination of underground water reservoirs.*
- Product reformulation/ Toxic and hazardous material use reduction:
 - *Example: Completely eliminate toxic substances such as brominated flame retardants in plastic materials*
- Water-based technologies:
 - *Example: using water based paints rather than solvent based paints (Volkswagen)*



Technique: B.3 Design for Benign Waste Disposition

- Treatment of waste streams to reduce environmental harm on disposal
- Waste sequestration/ waste isolation
 - *Example: Isolation of radioactive nuclear wastes in sealed containers*
- Ecosystem adsorption
 - *Example: fertilizer runoff can cause oxygen depletion in water bodies. One way to combat this problem is to build riparian buffer zones that act as natural filters along waterways.*
- Biodegradation
 - *Example: packaging in paper or biodegradable plastics*



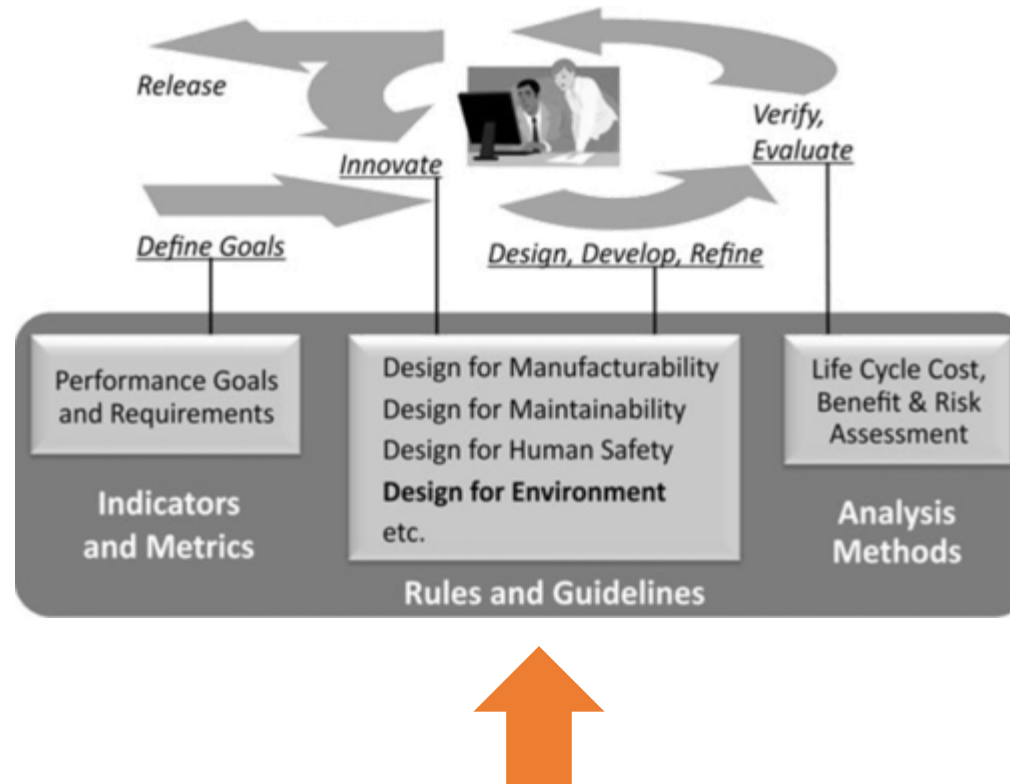
1.6 Design for Revalorisation and Capital Protection & Renewal

Dr Tim Walmsley

Recap: The Seven Principles of DFE

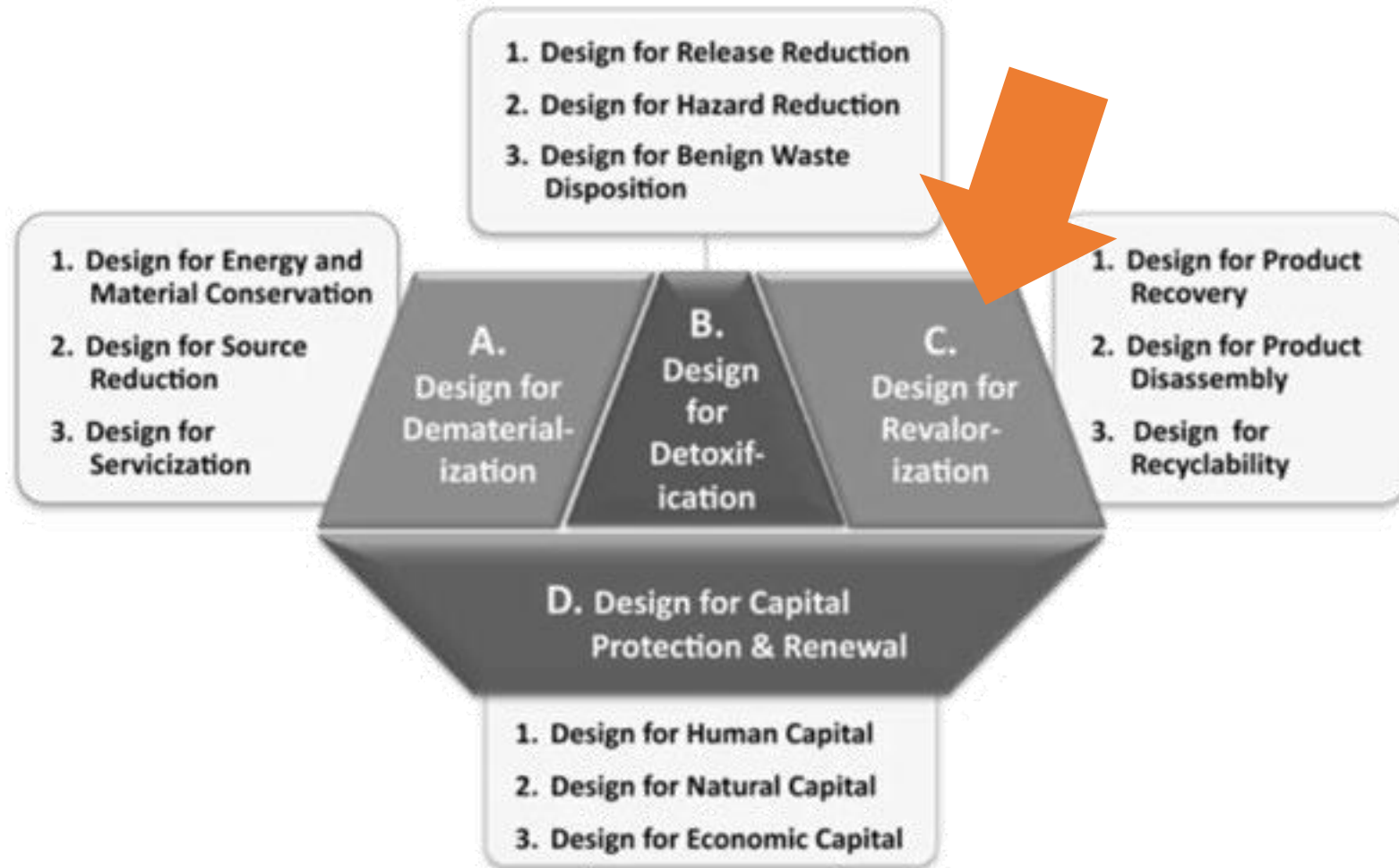
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Second stage of product development...



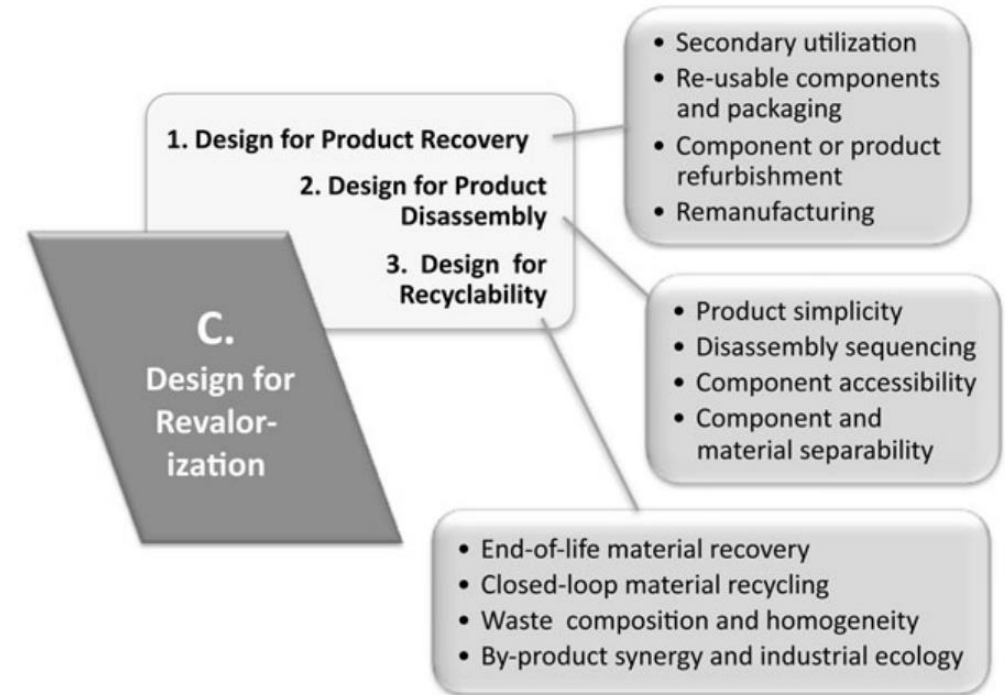


Design for revalorization...



Design for revalorization

- **Revalorization**
definition: methods for recovering value from a product at the end of its useful life
- The fundamental purpose of revalorization is to prevent a product or material from becoming a waste by diverting it to an **economically viable use**

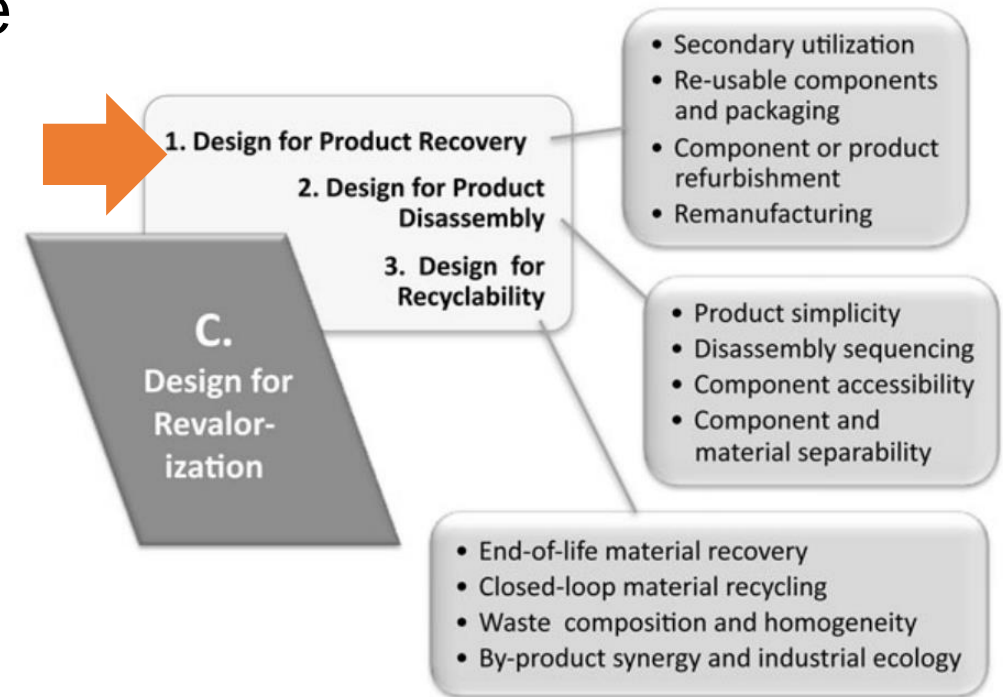


Technique: C.1 Design for Product Recovery

- Increasing end-of-life product value

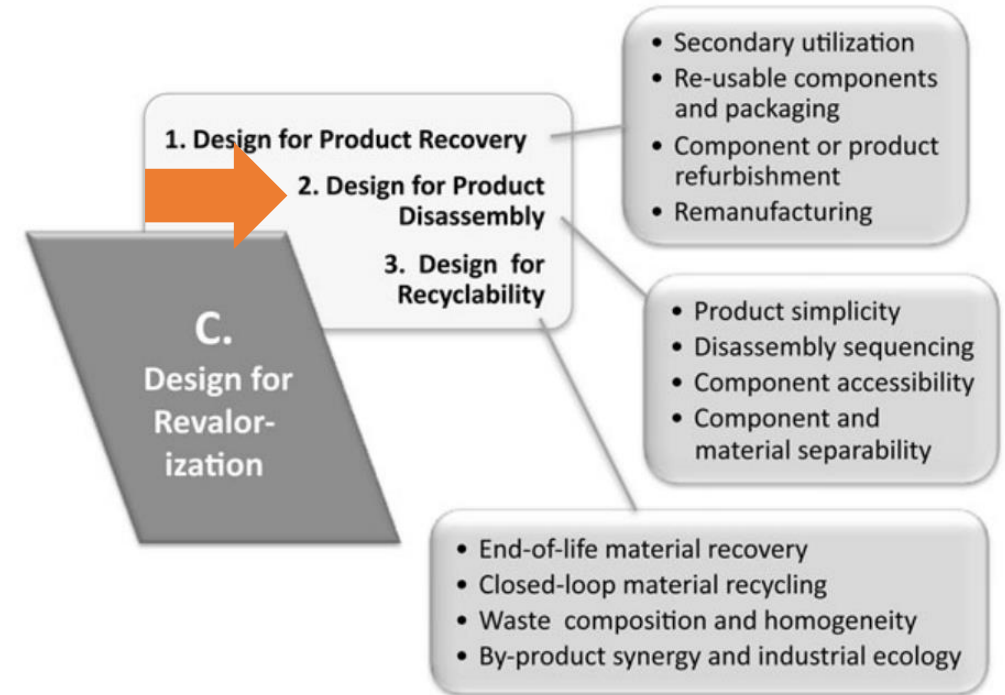
- Examples:

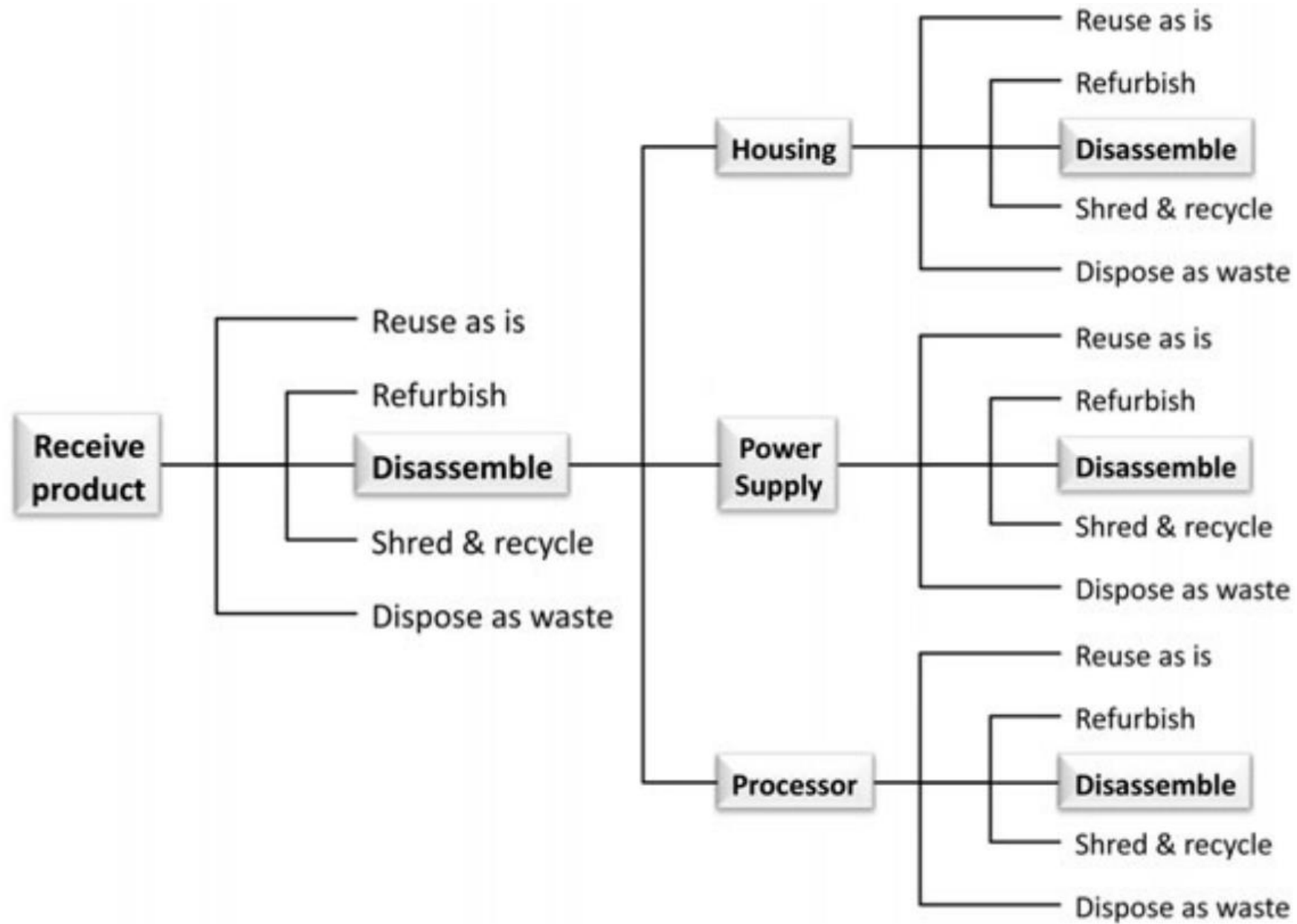
- *Closed loop remanufacturing*
- *Facilitate nondestructive removal of components*
- *Design reusable packaging*



Technique: C.2 Design for Product Disassembly

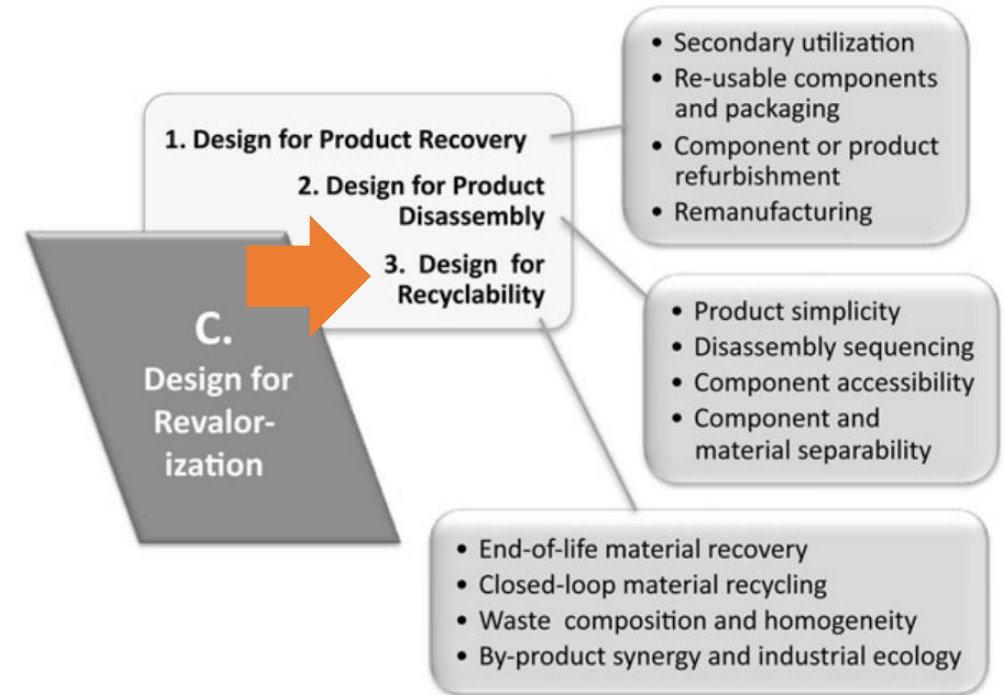
- This is an important prerequisite to other end-of-life considerations, such as component separability and recyclability.
- If you can't get access to a component, no matter how valuable, then you cannot recover it.
- Product Simplicity (fewer parts, fewer materials etc.)
- Disassembly Sequencing- displayed as a disassembly tree...





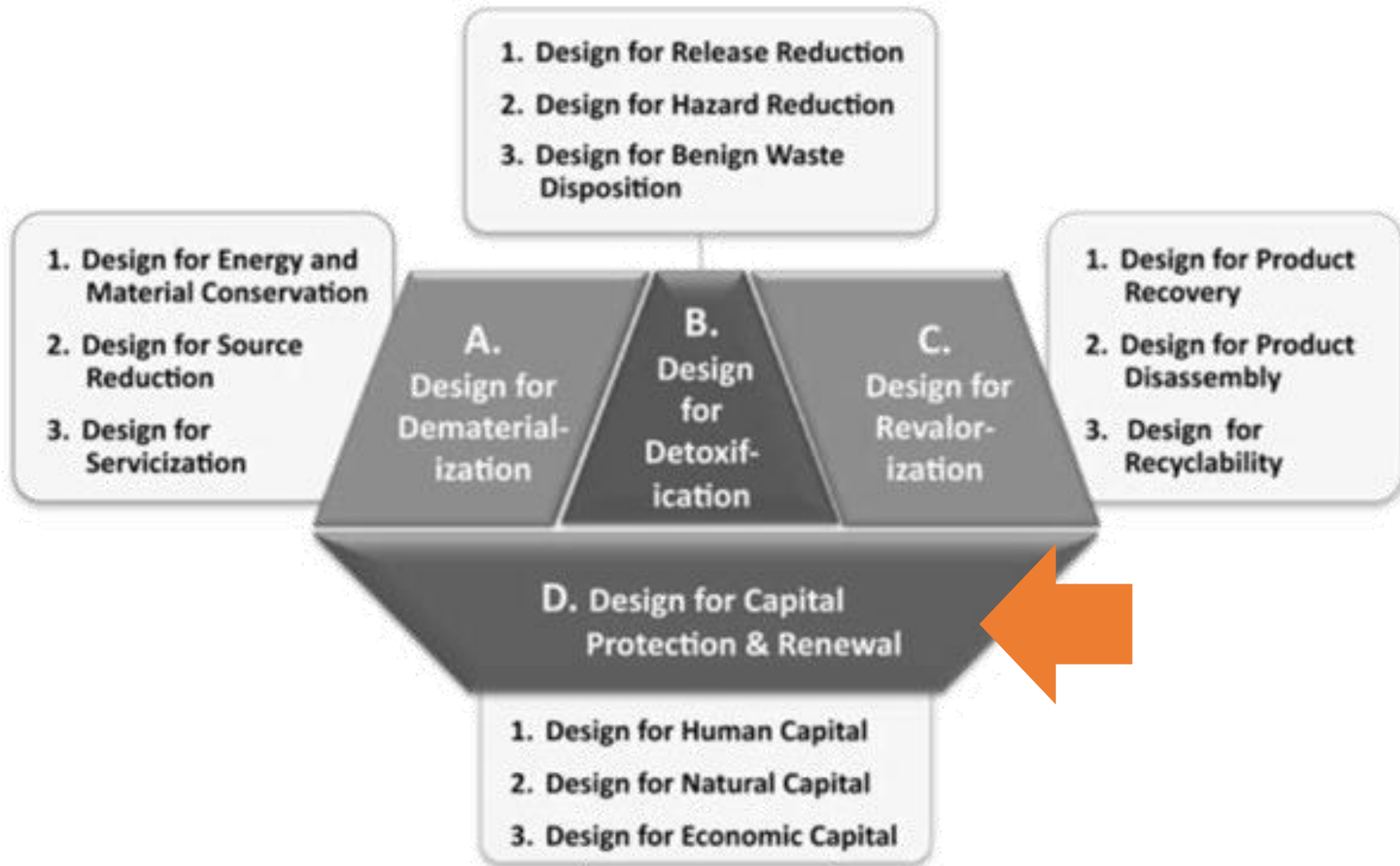
Technique: C.3 Design for Recyclability

- Zero waste goal not always practical
- Product materials or components that cannot be recovered economically become potential wastes.
- These can still be recycled and thus converted into useful by-products
 - *Example: Closed-Loop Material Recycling of aluminum cans.*
 - *Example: Bottled water using recyclable materials which are marked for recycling/ easy to separate from the rest of the product (ideally no stickers).*
 - *Example: Biomass wastes (e.g., wood chips) designed for waste incineration (the last resort) in waste-to-energy facilities*



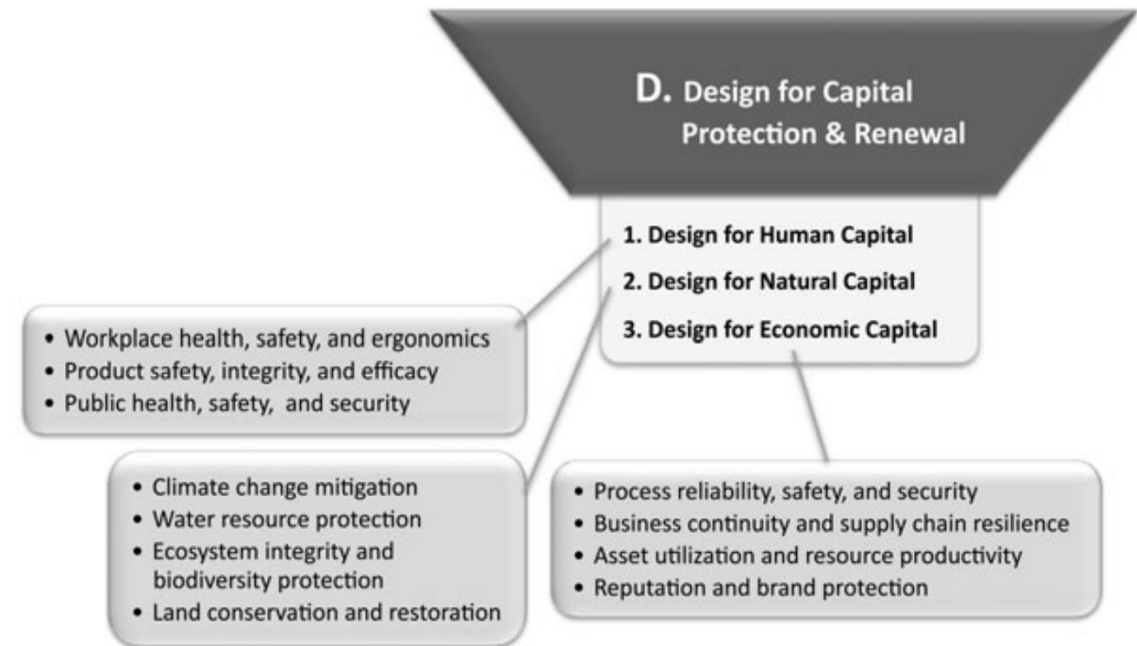
Design for Capital Protection and Renewal...





Design for Capital Protection and Renewal

- **Capital includes** human, natural, and economic capital.
- **Protection** refers to the maintaining of capital
- **Renewal** involves the restoring, reinvesting, or new capital generation.





Technique: D.1 Design for Human Capital

- **Human Capital:** skills and knowledge of management and employees.
- It is part of intellectual capital (includes patents and proprietary knowledge, relationships with customers and suppliers)
- Workplace Health, Safety, and Ergonomics
 - *Example: Use of Material Safety Data Sheets (MSDS)*
 - *Example: Ergonomic design of processes that prevent chronic injuries associated with lifting, bending, or other repetitive physical activities*
- Product Safety, Integrity, and Efficacy
 - *Example: screen / audit suppliers to prevent contamination risks (e.g. melamine from China)*
 - *Example: Prevent product abuse or misuse with clear warning labels, child proof lids etc*
- Public Health, Safety, and Security
 - *Example: fenced off areas where heavy machinery is being operated*

Technique: D.2 Design for Natural Capital

- **Natural Capital:** the ecological resources and services that make possible all economic activity, indeed all life
- Ecological resources include edible organisms, sand, wood, grass, metals, and minerals. Ecological services include various forms of energy provided by the water cycle, wind, tides, soil, and pollination
- Most ecosystem services have been degraded due to unconstrained economic development.



Examples: Design for Natural Capital

- **Energy conservation:** Conserve or substitute renewable energy sources (e.g. hydro or wind power) for non-renewable (fossil fuels).
- **GHG emissions reduction:** Sources of GHG emissions include fossil fuel combustion, industrial processes (e.g. cement calcination), agricultural processes (methane from cows), improper waste disposal (e.g. methane pockets in landfill from organic material)
- **Water resource protection:** could address through closed-loop recycling of process water, use of automatic shutoff valves and flow reducers, Utilize non-potable water for industrial operations to reduce demand on municipal water supplies
- **Ecosystem Integrity, Biodiversity, and Land Conservation:** Harvest crops without exceeding the capacity of the ecosystem. Restore damaged resources by reintroducing native plant and animal species.

Technique: D.3 Design for Economic Capital

- **Economic Capital:** productive assets of a company. Includes facilities, equipment, working capital.
- Many companies have sought to improve their asset utilization by identifying underutilized assets, such as idle equipment, and recovering their value



Examples: Design for Economic Capital



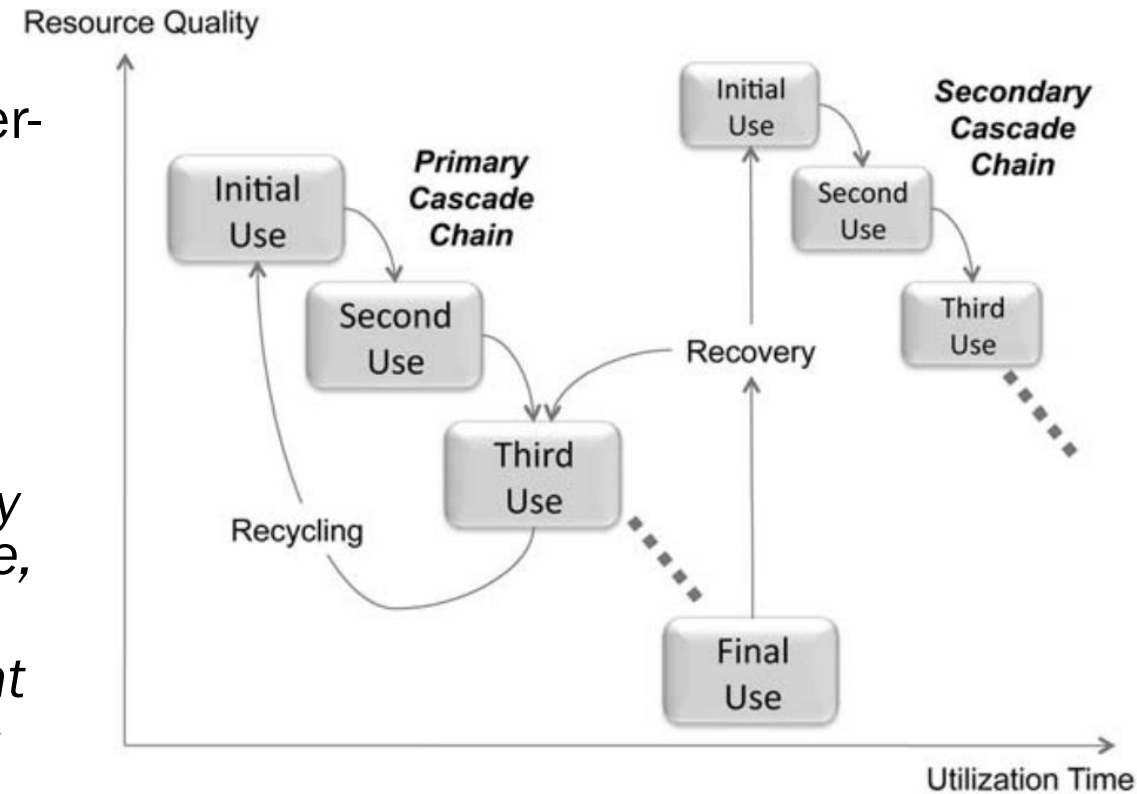
- **Process Reliability, Safety, and Security:** one of the most important drivers of plant availability. A minor safety incident, even if it causes little or no damage / injury, can disrupt operations and cause delays
 - *Example: minimize storage of caustic or flammable materials and use of high-temperature or high-pressure vessels.*
- **Business Continuity and Supply Chain Resilience:** the ability of an organization to maintain operations despite disruptive events
 - *Example: risk of supply chain disruptions may be higher when processes are sensitive to variations in climate conditions, constrained by the availability of scarce natural resources, dependent on a continuous energy supply*

Examples: Design for Economic Capital

- **Asset Utilization and Resource Productivity:** resource productivity can be measured in terms of annual production per dollar of invested capital (linked with eco-efficiency). “doing more with less”
 - *Example using the same equipment for a range of processing lines- Meat works use lamb processing line equipment for bobby calves as well.*
- **Reputation and Brand Protection:** Can account for up to 90% of a firm’s market value.
 - *Example: Use of renewable energy systems such as wind/solar power.*
 - *Many companies have succeeded in linking their brand with an “eco-friendly” image, improving consumer awareness and customer loyalty. However, if such marketing campaigns are not backed up with a genuine enterprise-wide commitment, they can be viewed as “greenwashing.”*

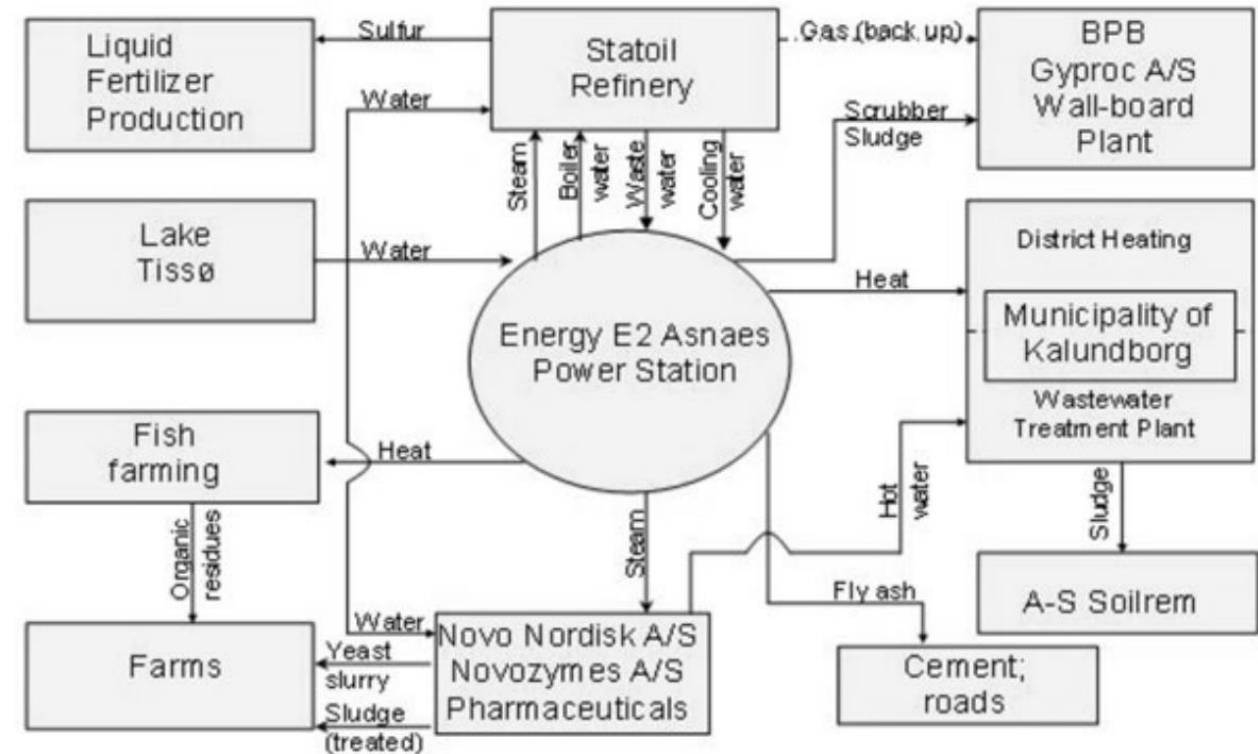
Examples: Design for Economic Capital

- **Resource Cascading:** a sequence of resource uses that extracts as much economic value as possible as resource evolves from higher quality to lower-quality forms
 - ***Example:** solvent used for degreasing in electronics manufacturing*
 - *Transferred to a metal cleaning operation.*
 - *Used in repeated cycles until eventually being relegated to its lowest quality use, as a paint solvent.*
 - *Purification and recycling, shifts solvent back up the cascade chain to a higher-quality use.*



Larger scale option: Industrial Ecology

- Shifting from the traditional linear model of industrial systems to a closed loop model that resembles the cyclical flows of natural ecosystem
- One facility's wastes can become another facility's feedstocks, leading to financial gains for both parties.
- The conversion of industrial process wastes to by-products, called by-product synergy (BPS).



Summary: Guideline Strategies

